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Impact of basic artificial intelligence (AI) course on understanding concepts, literacy, and empowerment in the field of AI among students

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Abstract

With the development of information technologies and information processing methods, it is important to provide high-quality education in the field of artificial intelligence (AI). The study aims to investigate the impact of an educational course on AI on the comprehension of concepts, literacy, and empowerment in the field of AI among students of higher educational institutions. The experiment involved 125 students from Hohai University in China. As a result of taking the training course, students were able to improve their understanding of concepts (increasing their average score from 6.33 to 9.69), literacy (from 2.94 to 3.99), and empowerment (from 3.90 to 4.04) in AI. The resulting data statistically confirmed the effectiveness of the developed course for improving confidence in the field of AI. The training module can be applied to improve confidence in the field of AI for students in various careers, as information competence is important these days and increases the success of graduates in employment. When it comes to further research, the encouraging results of this study suggest opportunities for promoting this training program among a diverse group of participants. To confirm the effectiveness of the developed course, it can be conducted among students in schools and other educational institutions, reducing it to even more basic if necessary.

KEYWORDS

digital technology, information literacy, information technology, learning artificial intelligence

1 | INTRODUCTION

Artificial Intelligence (AI) defines a new era in technology, transforming the way people interact with information and address complex tasks [7]. However, as these technologies are directly integrated into everyday life, challenges arise in understanding and effectively harnessing their potential, particularly concerning the younger generation shaping the future of this technologically enriched era [8].

Therefore, investigating the impact of an educational course on AI on the comprehension of concepts, literacy, and the expansion of AI capabilities among students is pertinent and strategically crucial. With the increasing significance of AI in various fields, the ability of young thinkers to grasp the fundamentals of this science not only determines their competitiveness in the job market but also shapes a new wave of innovation and development [13].

In this article, an analysis will be conducted on the impact of a foundational course in AI on students' comprehension of concepts, literacy, and empowerment in the field of AI. Within the context of AI, the understanding of concepts, literacy, and capability expansion can be defined as follows. Understanding AI concepts involves students' ability to comprehend the fundamental principles and theoretical concepts underlying AI, encompassing an understanding of key notions such as machine learning, neural networks, natural language processing, and other pivotal aspects [9]. It also involves the capability to apply acquired knowledge in real-world scenarios and tasks aimed at problem-solving and creating new products or solutions utilizing AI. There is also a discussion about the role of the ability to develop and implement algorithms. For example, some researchers do not include this ability in the core competencies of AI literacy [22]. They believe that AI literacy is a more basic competence that remains at the level of "Competency 17 (Programmability) understand that agents are programmable"; literacy also entails an understanding of ethical dimensions and responsible use of AI, addressing challenges related to privacy, transparency, and societal impact [22]. Empowerment in the field of AI signifies that students can enhance their skills by participating in practical projects and competitions where they apply their knowledge to real-world tasks [24]. This encompasses the ability to pursue individual research directions in AI, contributing to the expansion of knowledge and innovation and engaging with communities such as open sources, industrial forums, and working groups for continuous improvement and broadening of capabilities. Understanding concepts, literacy, and empowerment in AI are pivotal elements that assist students in adapting to the rapidly changing world of technology and effectively utilizing AI to address contemporary challenges [31]. The results of the previous study demonstrated that literacy courses successfully prepared participants for conceptual understanding of AI, and participants felt empowered due to significant improvements in their literacy and conceptual comprehension of AI [14].

The primary objective of the research is to investigate the impact of an educational course on AI on students' comprehension of concepts, literacy, and empowerment in the field of AI among students of higher educational institutions. Through this study, we aim to unveil the potential of education in AI in shaping a new generation of leaders and innovators. The tasks included:

- develop a basic training course on AI,
- determine students' literacy, level of understanding of concepts, and empowerment of AI before the experiment,
- assess the participants' competence, level of understanding of concepts, and empowerment of AI after the experiential learning,
- examine participants' perceptions of their learning achievements after completing the course and conclude the effectiveness/ineffectiveness of the developed course.

Research questions:

1. Can the developed course improve college students' understanding of AI concepts?
2. Will students' AI literacy levels increase after taking the course?
3. Will the course have an impact on improving students' knowledge of AI empowerment?

2 | LITERATURE REVIEW

2.1 | AI literacy courses

Recent research in the field of AI literacy indicates a growing interest in issues related to education and proficiency assessment in this domain, with many countries actively implementing AI education programs [16, 19, 35]. Studies in this direction aim to identify the most effective methods and strategies enabling students to acquire both conceptual understanding and practical skills in AI. Adaptive educational programs have become a crucial component, considering the individual needs and preparedness levels of each learner [27]. The development of pedagogical methods and tools, such as online courses and interactive lessons, is also being explored to ensure quality and accessible AI literacy education [5, 40, 41]. Previous research on different approaches to AI literacy and basic AI courses suggests that this direction of research is relatively new. Therefore, research on AI literacy in higher education institutions may lag behind similar research in other sectors. Most studies focus on courses aimed at developing AI literacy. Nevertheless, it is still challenging for many to structure such courses or select content for them [18]. Thus, it is necessary to compare different educational frameworks for the creation of new AI courses. Previous studies have emphasized the importance of focusing on the representation of key competencies that underpin AI literacy. However, there is a lack of agreement regarding which competencies to consider as key and their definitions differ between different studies. Studies also call for the development of adequate educational programs so that everyone can take advantage of AI technologies [16]. The analysis of courses aimed at AI literacy indicates that the combination of theoretical material and practical tasks is an effective approach to teaching AI competencies at higher educational institutions [22].

2.2 | Measuring AI literacy

Simultaneously, research is underway to understand and enhance the assessment of AI literacy. Defining metrics and standards for AI literacy is a crucial task, as standardization of measurement plays a key role in comparing levels of education and skills among students [23]. Recent developments in the assessment of AI literacy, within the context of existing tools, represent a significant step toward expanding the measurement tools for users' competence in AI utilization. Several validated instruments, such as the Meta AI Literacy Scale [2], assessments of attitudes toward AI [33], and user competence measurement tools [17], among others [20, 32, 37], have been established. However, there are conflicts in the theory and methodology of AI literacy assessment. Some studies emphasize technical programming skills [22], while others tend to define literacy as the ability to understand and apply machine learning algorithms [29]. Contradictions in approaches to measuring AI literacy lead to gaps and limitations in existing research. The lack of a systematic definition of literacy and the absence of universal standards complicate result comparisons and understanding of the level of student preparedness. Therefore, it is crucial to address and resolve identified conflicts and gaps in AI literacy research.

2.3 | Research contributions and potential issues in AI utilization

This study included the development of evaluation tools to test the effectiveness of a basic AI course. The tools evaluate both the conceptual aspects, as well as the understanding and application of AI. The instruments developed in this research focus on understanding concepts, literacy, and expanding capabilities in the use of AI among students. The contextualization of the instruments considers the results of previous research and the evolution of the field of AI literacy. The new tool can contribute to defining the level of literacy and improving assessment methods in this field. The novelty of the research lies in the implementation of a comprehensive approach that encompasses both technical and conceptual aspects of AI literacy. It is essential to identify and resolve conflicts, broaden the scientific discourse, and make a significant contribution to understanding AI literacy in the context of the modern educational paradigm. In the age of technological innovations, the instruction of a foundational course in AI emerges as pivotal for students, offering a range of positive aspects. First, such a course fosters a profound understanding of AI concepts, wherein students grasp not only the technical aspects but also the principles guiding the functioning of algorithms and AI models [22]. Second, AI

education enhances students' literacy in this field, equipping them with practical programming skills, data manipulation abilities, and the utilization of relevant algorithms—an essential component of modern education [3]. Third, the course broadens students' opportunities, unveiling new perspectives in various life domains, such as business, science, and medicine, facilitating the easier implementation of technological solutions, and contributing to their professional development [8]. Finally, AI education prepares students for future employment, where technology becomes a vital component, providing them with a competitive edge in the job market by possessing contemporary technical skills [8]. Despite the positive aspects, there are potential negatives that warrant consideration. Ethical issues may arise in AI education, particularly concerning the use of AI in domains where personal data is involved, and critical decisions are made [39]. There is a risk of dependence on AI technologies that may overshadow critical thinking and a creative approach to problem-solving [11]. Lastly, the fear of potential job displacement due to automation may instill a certain negative sentiment among students [12]. It is crucial to strike a balance between the development of technical skills and an understanding of the ethical, social, and economic aspects of AI use to ensure a comprehensive approach to education.

3 | METHODS AND RESOURCES

3.1 | Study design

To achieve the research objectives, a basic AI course was developed and implemented in experiential learning. To assess the contextualized impact of the AI course, participants in the study underwent mandatory tests both before and after completing the instruction. The evaluation encompassed a comparison of the knowledge and skills of participants before and after the course, specifically assessing their abilities to understand concepts and apply skills acquired during the AI course. This approach facilitated a contextualized assessment and identified the specific impact of the education on the change in the participant's level of knowledge and understanding.

3.2 | Respondents

A total of 125 students from Hohai University in China were selected to participate in the experiment. Of these, 60 were women and 65 were men, that is, approx. equal sex ratio. The participants' ages ranged from 18 to 20 years old. They are all Chinese by nationality. It is also noteworthy that all participants in the study represented

a homogeneous study population, further standardizing the experimental conditions. The students participated in the research during the second semester of their university education. The research participants undertook an elective course in AI and received academic credits upon successful completion of the course. The students had not previously taken any training in AI but naturally had some digital skills as members of today's generation. To simplify the study, the participants were divided into three groups: 40 students in the first and second groups and 45 in the third group (random assignment). There were no differences between the groups; the division into three groups was done because it would have been difficult for one teacher to teach 125 at a time, and other teachers were not involved because different teaching styles might then have affected the study results.

3.3 | Program content

The course syllabus was developed by an information technology professor who studied AI for the past 12 years and has been teaching AI to students at various levels for 7 years. Within the course, seven sessions were developed on the following topics (the main issues that were addressed are shown in parentheses):

1. General provisions on AI systems (The concept of intelligence. The concept of AI. The concept of data and knowledge. Intellectual tasks. Various interpretations of intellectual activity. Areas of application of AI. Perception and pattern recognition. Logic games. Expert systems. Classification of AI systems).
2. The artificial neural network (Structure and properties of an artificial neuron. Classification of neural networks and their properties. The topology of neural networks. Simple single-layer neural networks. The process of learning neural networks. Application of a trained neural network. Basic feature sets of neural network modeling programs. The use of artificial neural networks).
3. Multilayer neural networks (Topology of multilayer neural networks. Network learning algorithms. Multilayer perceptrons [MLP]. The backpropagation algorithm. Criteria for the adequacy and evaluation of MLP models. Using MLP models to solve management problems).
4. Using AI to solve problems of business intelligence systems. Self-organizing neural networks (Competitive networks. Kohonen map [Kohonen network]. The principle of Kohonen maps. Teaching the Kohonen network. Solving classification problems based on the Kohonen network).
5. Hopfield networks and associative networks (The architecture of Hopfield networks. The principle of Hopfield networks. The Hopfield network learning algorithm. Classification of associative networks. Bidirectional associative memory [BAM] networks. The principle of the BAM network. The BAM network learning algorithm. Identification problem solving).
6. Genetic algorithms (Genetic algorithms. Basic concepts used in genetic algorithms. Genetic operators: crossover, mutation, etc. Applications, disadvantages, and advantages of genetic algorithms. Intellectualization of modeling techniques using genetic algorithms).
7. AI system for data retrieval and TextMining (The structure and operation of search engines and systems. Zipf's law. Intelligent agents. Parsing and indexing system. Principles and methods for ranking search results. Semantic networks. Data description and semantic network standards).

The course's main objectives are to learn theory and skills to develop a basic understanding of the application of AI systems, as well as to acquire the abilities and skills to solve problems using AI systems and learn the theory and skills of creation and application of AI systems and expert systems. The tasks also involve the study of approaches to processing and representation of knowledge in intelligent systems.

Upon completion of the course, students should be able to:

- use AI systems to solve applied problems in different subject areas,
- design AI systems, expert systems, knowledge bases,
- apply methods for solving problems in weakly formalized industries,
- use different methods of teaching.

The items presented above are not course objectives; they are simply competencies that students should acquire after learning. The acquisition of these competencies was not measured; it was measured in terms of literacy, conceptual understanding, and empowerment in the field of AI.

3.4 | Implementation and measurement

Students were taught face-to-face by one instructor (course creator) in all groups. The entire training process took 4 months, and a total of 14 sessions were conducted: seven lectures and seven practical sessions. The duration of the class was 90 min. All participants attended lecture classes together, and for convenience, they attended

practical training in groups. During the theoretical lecture sessions, the instructor employed various pedagogical methods to ensure the maximum effectiveness of students' comprehension of the material. The presentation of information, illustrations and examples, discussions, and the use of slides and multimedia resources, along with practical tasks, aided students in a deeper understanding of the concepts. Practical sessions were conducted in the form of group work, where students applied their knowledge to solve specific tasks. The application of the case method, group exercises, independent practical tasks, and the discussion of results contributed to the development of students' communication and analytical skills.

Instrumental measurements before and after the intervention were performed using the same procedure. Participants in the classroom took the three tests described below, which were uploaded to Google Forms for convenience. A smart device with an Internet connection was required to pass the tests. 30 min were allocated for filling out. Figure 1 shows a brief diagram of the experiment.

Students' perspectives were collected through in-depth, semi-structured interviews that averaged 1 h in length. After initial contact, an invitation explaining the goals and objectives of the study was sent in writing via a personal Facebook Messenger message. Subsequently, an interview was scheduled with each student at a mutually convenient date and time. A semi-structured questionnaire was used to facilitate discussion. Additional questions were also used as needed. Interviews continued until data collection no longer revealed new patterns or themes.

3.5 | Evaluation instruments

To assess the effectiveness of an experimental course on improving students' AI confidence, relevant literature was used to develop tests on concepts, literacy, and empowerment in AI [10, 15, 22, 28, 36]. The following steps were taken in the development of the tools:

- literature review to generate initial test ideas,
- drafting of questionnaire items,
- refining and adjusting the instruments as needed,
- testing the instruments for reliability.

In the concept test, there were 14 multiple-choice questions to be answered (Appendix 1). These questions addressed the AI concepts covered in the course and were designed to assess participants' progress in understanding these concepts. The conceptual test was

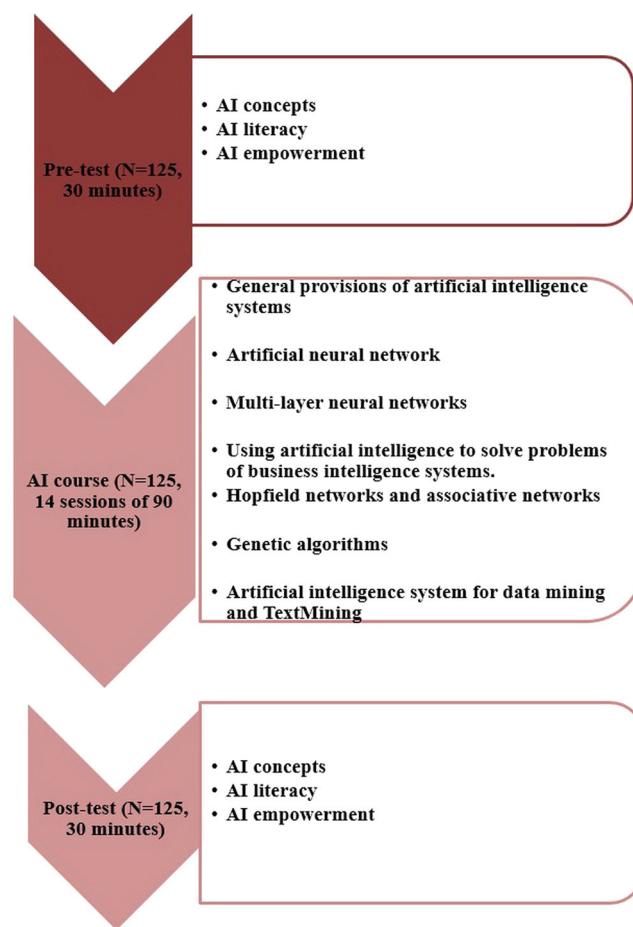


FIGURE 1 Experiment.

designed to assess the tangible demonstrations of students' competence. The maximum possible score for the test is 14 points.

The literacy test (Appendix 2) assessed participants' perceived level of AI literacy, not their actual level of AI literacy (i.e., it was a self-report rather than an objective measure). Students had to agree/disagree with each item on a 5-point Likert scale (1–strongly disagree, 5–strongly agree). For example, one of the items was “I can explain the use of AI in various fields.” There was a total of 15 items. The literacy questionnaire aimed to assess anticipated levels of competence.

The empowerment test was designed to identify skills in dealing with AI and contained 17 items to be assessed on a Likert scale as well (Appendix 3). The questionnaire contained questions about students' perceptions of their levels of empowerment to technology. Students had to agree/disagree with each item on a 5-point Likert scale (1–strongly disagree, 5–strongly agree). Sample question: “I believe that AI knowledge is significant for my education and professional development.” The instrument was reliable as Cronbach's alpha for the concepts,

literacy, and empowerment tests was 0.67, 0.88, and 0.95, respectively. Additionally, a semi-structured interview questionnaire was developed to gather qualitative data (Appendix 4). A total of 125 interviews were conducted.

3.6 | Data analysis

The study utilized a paired *t*-test to analyze data from various tests (concept, literacy, and empowerment tests) and determined mean values and standard deviations. Thematic analysis was used to analyze semi-structured interviews. Preliminary analysis included interview transcripts and multiple readings of the transcripts. Subsequent analysis included identifying emerging themes, grouping them into categories, and creating summaries of student narratives. Color coding was used to facilitate the study, and refinements were made at each coding step. During thematic analysis, similar codes were grouped to form thematic categories, and prominent patterns and relationships between themes were identified. While quantitative data from student participants were analyzed using SPSS software (version 28), NVivo software was used to analyze qualitative data.

4 | RESULTS

4.1 | Advances in understanding AI concepts

The results of the concept test before and after the experimental training are shown in Table 1. The data of the paired *t*-test can also be found there.

Students made progress in their understanding of AI concepts, as evidenced by an increase in their mean score from 6.33 to 9.69. The effect size (η^2) was 0.60, meaning a small to medium effect.

4.2 | Improving AI literacy

Literacy test results before and after the experimental training and paired *t*-test data are shown in Table 2.

Participants improved their AI literacy as test scores increased from 2.94 to 3.99. The effect size (η^2) was 0.62, meaning a small to medium effect.

4.3 | AI empowerment

Pre- and post-experimental learning empowerment test results and paired *t*-test data can be found in Table 3.

There was an improvement in the average score from 3.90 to 4.04, which was confirmed statistically. The effect size (η^2) was 0.63, meaning a small to medium effect. It can be concluded that the experimental training module increased students' sense of competence in the field of AI. The interview results also indicate that the AI skill empowerment course was effective. Table 4 notes students' perceptions of their accomplishments after the experiential training.

Students rated the course's practical usefulness for their professional and personal development at a high level. The majority of the participants interviewed stated that the experiential learning module helped them understand teaching strategies with AI technology. It was also noted that the course helped them to find topics for new learning projects and to find AI tools that can assess and monitor student learning. There were reports on how to improve the effectiveness of the teaching process. Students began to think more often about technology-related applications and their reliability for use in everyday life. This kind of response indicated that they showed a willingness to apply their knowledge in real-life situations. The creative self-efficacy of the participants developed.

5 | DISCUSSION

The results of the study indicate a significant impact of the AI educational course on students of higher educational institutions. Further discussion of the results in accordance with the questions raised in the study provides the following. The developed course considerably improved the understanding of AI concepts among the students. Thus, the average score on understanding concepts increased from 6.33 to 9.69, suggesting the

TABLE 1 Analysis of data from the pre- and post-test of AI concepts.

Test on AI concepts	Pre-test		Post-test		Paired <i>t</i> -test results	Effect size (η^2)
	Mean value	Standard deviation	Mean value	Standard deviation		
	6.33	2.07	9.69	2.61	13.91***	0.60

Note: $N = 125$.

*** $p < 0.001$.

TABLE 2 Analysis of data from the pre- and post-test of AI literacy.

Literacy in AI	Pre-test		Post-test		Paired <i>t</i> -test results	Effect size (η^2)
	Mean value	Standard deviation	Mean value	Standard deviation		
	2.94	0.63	3.99	0.47	16.01***	0.62

Note: $N = 125$.

*** $p < 0.001$.

TABLE 3 Analysis of pre- and post-test data of AI empowerment.

AI enhancement	Pre-test		Post-test		Paired <i>t</i> -test results	Effect size (η^2)
	Mean value	Standard deviation	Mean value	Standard deviation		
	3.90	0.44	4.04	0.46	2.97**	0.63

Note: $N = 125$.

** $p < 0.01$.

TABLE 4 Students' perceptions of their achievements after the experimental training.

Question	Descriptions	Participant responses
1. Application in education	• Ideas for learning projects;	After taking the course, I have ideas for new projects that are related to artificial intelligence (AI). I plan to continue my studies in the field of information technology. My main advantage will be the acquired knowledge in the field of AI. It is likely that the technology can find application in teaching. Learning English with the help of AI is a promising field of application. I think the acquired knowledge will help me in my work because I received a lot of resources for organizing activities in the field of the studied technology. Now when using digital technology, I wonder if AI applies there.
2. Application in work	• Continuing education;	
3. Use in daily life	• Enhancement of teaching; • Provision and improvement of instructional resources; • Reflecting on technology.	

success of the educational process. After completing the course, the students substantially increased their level of AI literacy. The increase in the literacy level from 2.94 to 3.99 indicates that the students gained more knowledge and skills in this field. The results also showed that the course improved and expanded the knowledge of students regarding the capabilities of AI. Although the changes in this indicator were not as significant as in understanding concepts and literacy, they were still noticeable (from 3.90 to 4.04). This result also indicates a positive impact of the course on the knowledge of students.

The obtained results exhibit similarities with certain aspects of other research but also reveal distinctive features. For instance, in comparison to prior studies where some uncertainty among students was identified, the well-structured course led to a significant increase in understanding key concepts and the development of AI literacy [4, 6, 21]. These results can be elucidated through the lens of several scientific

theories and concepts that define the interaction of students with AI. One of the key theories is the cognitive constructivism theory, which recognizes the active involvement of students in constructing their knowledge; learning through practical experience offered in the course could contribute to a deeper assimilation of AI concepts [26, 34, 38]. Additionally, the results can be viewed through the lens of the sociocultural approach theory, where the interaction of students with AI is considered an essential element of the sociocultural context [25]. The course might have facilitated active interaction and knowledge exchange among students, fostering a deeper understanding of the material and the development of AI literacy through a social context. In analyzing the results, the constructivism theory, emphasizing the individual nature of the learning process and the importance of student activity in mastering new concepts, should also be considered [1, 30]. The developed course, by emphasizing practical tasks and

interaction, allowed students to construct their understanding of AI, potentially influencing the development of literacy in this field.

In the context of various tests conducted in the study, it is worthwhile to discuss the relationship between self-assessment of competence and actual competence of students in different aspects, such as understanding of concepts, literacy, and the expansion of AI capabilities. If students highly rated their understanding of AI concepts, and the actual level of understanding is confirmed by the tests, it can be considered that the self-assessment was accurate and reliable. However, any discrepancies may indicate the need to enhance the objectivity of self-assessment. Similarly, if students highly assessed their AI literacy, and test results indicate a high level of literacy, it corroborates the credibility of self-assessment. In cases of disparities, factors influencing the perception of their skills can be investigated.

If the test on AI capabilities reflects positive development in students, it may indicate the effectiveness of the course in preparing them for the use of AI technologies. Analyzing the difference between self-assessment and actual achievements can reveal how accurately students assess their abilities. Regarding competence and perceived competence, if students corroborate their competence with actual achievements, it can be argued that their self-assessment reflects the real level of competence. In cases of discrepancies, this may indicate a tendency to overestimate or underestimate their abilities. Continuing research in this direction, relying on the results of various tests and analyzing differences between self-assessment and objectively measured indicators of competence in students, is crucial.

If necessary, this course can be adjusted to a more foundational level, considering the needs of beginners. Primarily, it is advisable to simplify the terminology, avoiding complex technical terms and excessive mathematical content. The course should commence with general principles of AI systems, focusing on basic concepts and application areas. Over time, a step-by-step introduction of concepts is recommended, with each lesson addressing a specific aspect of AI, progressing from simple to more complex topics. Providing students with clear examples and highlighting the practical applications of AI technologies in real life is crucial. To facilitate comprehension, concrete examples and applications of AI in various domains, such as business analytics, data search, and TextMining, can be employed. Additionally, reducing mathematical complexity and emphasizing fundamental principles is recommended. The introduction of practical tasks and projects would assist students in applying acquired knowledge in

practice. This approach would contribute to an effective understanding of AI at a basic level.

The obtained research results attest to the potential and effectiveness of the developed course, which can serve as a prototype for further educational programs in the field of AI. It is notable for its distinctive approach, setting it apart from other similar educational initiatives. However, for a detailed exploration and clarification of these results, it is recommended to conduct additional research that can identify the role and level of influence of various predictors of success in the AI course. These predictors may include teaching style, motivation, self-efficacy, or other unidentified factors. Additionally, it is advisable to adjust teaching strategies for different groups in the practical component and observe whether differences emerge among these groups. Furthermore, a comparison between self-assessment and actual competence evaluation, such as exam performance, could provide valuable insights.

5.1 | Limitations

This study is limited by the following factors. Students from the same institution participated. The Cronbach's alpha coefficient for the concept test was 0.67. The acceptable limit should be 0.6 to 0.7, especially for new instruments. It is still important to improve the reliability of this instrument. Simultaneously, the high-alpha coefficient (0.95) for the Empowerment test may suggest that all questions in the test could be interchangeable or that the test might include redundant aspects. Excessively high-internal consistency may indicate that all questions measure the same aspect of the concept, which, in turn, could lead to excessive homogeneity and limit internal validity. For further refinement of the research instrument, it is crucial to scrutinize each question of the test and determine its specific contribution to measuring the concepts. In future research, alternative methods for assessing internal consistency, such as split-half testing or factor analysis, can be employed to obtain a more comprehensive understanding of the test's reliability and address potential shortcomings identified through Cronbach's alpha coefficient. Another study disadvantage is the lack of a control group. In this study, all data were based on the self-reports of individual respondents. This could have led to an overestimation or underestimation of the results due to general method variance. In other words, individual differences between respondents could have affected the results of the study, especially if the self-reports were not equally accurate or subjective. In future studies, the use of more objective indicators that do not depend only on

self-reports will help to remove these concerns. Due to the above limitations, all conclusions drawn from this study remain preliminary but represent an important starting point for further quantitative work in the field.

5.2 | Ethical aspects

The study observed the following ethical principles: voluntary participation, informed consent, anonymity, confidentiality, and informing about the results. Physical, social, psychological, and all other types of harm were reduced to an absolute minimum.

5.3 | Implications

The obtained results and conclusions from the research have significant implications for the educational technology and AI research community. Firstly, they underscore the success of the AI course in building students' trust in this field. Understanding concepts, improving literacy, and expanding AI capabilities became possible through effective teaching, which may prove beneficial for further applications in various professional domains. The integration of AI courses can significantly impact students' levels of understanding and literacy in this area. It is also crucial to note that students recognized the utility of practical aspects of the course, such as experiential learning and working with AI technologies, for their professional and personal development. Considering case adaptability, the expansion of the course to different participant groups, including students from various educational institutions and professionals from different fields, could be explored. Additional adaptations may involve optimizing test questions and assessment methods to enhance the reliability and objectivity of the study. It is also important to consider the possibility of introducing a control group to obtain comparative data and confirm the course's effectiveness.

In addition, this study has significant potential to contribute to the research community in the following aspects. These findings can help identify effective methods and approaches for teaching AI. The analysis and evaluation of the methods used in this course can provide valuable input for the further development of AI educational programs. Data on changes in understanding, literacy, and empowerment in AI after completing a course can be useful for comparing with other AI educational programs and determining their effectiveness. The results can provide justification for the development and implementation of new educational programs on AI at higher educational institutions. The findings can also

serve as a basis for changes in AI curricula and policies. Discovering success in AI training can encourage greater interest among students and teachers in the field. This interest can lead to greater demand for AI education programs and resources.

6 | CONCLUSIONS

This study developed and implemented an educational module designed to increase confidence in the field of AI. As a result, participants had significant improvements in their understanding of concepts, literacy, and AI, which was statistically significant. There is a very real and urgent need to prepare students for a future in which digital tools disrupt all sectors of society, even when considering only the existing technologies without innovations. However, the undifferentiated call for digital skills does not meet students' needs. The use of digital technology will affect some areas of learning more than others, and different technologies will be relevant to different groups of students. For this reason, greater precision is needed in the public discussion of digital skills, and educational institutions need to take a case-specific approach that allows them to implement policies that are appropriate for various user groups.

The developed course can be applied to improve confidence in the field of AI for students in various professional areas, as information competence development is essential these days and increases the success of graduates in the labor market. After collecting and analyzing data from the additional implementation of the curriculum, a model of AI literacy competencies is expected to be developed to further contribute to AI education. Research is needed on the application and improvement of AI and the possibility of extending human knowledge. In future studies, it is important to expand the sample to increase the scientific data in the study area.

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CONFLICT OF INTEREST STATEMENT

The authors have no conflict of interests to declare that are relevant to the content of this article.

DATA AVAILABILITY STATEMENT

All data generated or analyzed during this study are included in this published article. There is no additional data.

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APPENDIX 1

The concept test

1. What concept reflects the ability of artificial intelligence (AI) systems to use probabilistic methods to solve tasks?
 - a) Cybernetics
 - b) Bayesian statistics
 - c) Virtual reality

- d) Quantum computing
2. What is the main goal of AI systems?
 - a) Increasing the number of programs
 - b) Creating unparalleled computer games
 - c) Providing computers with the ability to learn and make decisions
3. What concept describes the ability of AI systems to learn from experience?
 - a) Impulsiveness
 - b) Heuristics
 - c) Neural networks
 - d) Machine learning
4. What are neural networks in the context of AI?
 - a) Powerful computer processors
 - b) Structures modeling the operation of the human brain
 - c) Special programs for image processing
 - d) Technical devices for virtual reality
5. What principle underlies the backpropagation algorithm in neural networks?
 - a) Finding the shortest paths
 - b) Transmitting signals from limbs to the brain
 - c) Approximation and error correction
 - d) Creating a network of genetic algorithms
6. What are the main tasks of expert systems?
 - a) Manufacturing high-quality products
 - b) Solving complex mathematical problems
 - c) Modeling expert knowledge and working with it
 - d) Generating and analyzing big data
7. What is the main idea of genetic algorithms?
 - a) Modeling the functioning of genes
 - b) Using elements of natural selection to find optimal solutions
 - c) Creating precise mathematical formulas
 - d) Using quantum computing technology
8. What task do TextMining systems solve?
 - a) Translating texts from one language to another
 - b) Analyzing and extracting information from text sources

- c) Creating textual advertising banners
- d) Creating stylized texts for artistic works

9. What is the name of the network architecture that studies patterns and properties of data?

- a) Kohonen network
- b) Hopfield neural network
- c) Multilayer neural network
- d) Associative memory system

10. How is the Internet of Things (IoT) related to AI systems?

- a) The IoT is the most crucial component of all AI systems
- b) AI systems interact with IoT devices, processing and analyzing their data
- c) The IoT exclusively uses AI algorithms
- d) These concepts are not related

11. Which AI systems are used for object classification and decision support?

- a) Self-organizing neural networks
- b) Expert systems
- c) Genetic algorithms
- d) Hopfield networks

12. How does AI impact the field of business analytics?

- a) Applying AI makes business analytics unnecessary
- b) AI helps automate data analysis and decision-making processes
- c) Business analytics and AI are independent of each other
- d) AI is exclusively used in the medical field

13. What aspects of using AI in business analytics can be problematic?

- a) Increasing efficiency and accuracy
- b) Creating autonomous systems
- c) Ethical and confidentiality issues
- d) Reducing the volume of data for processing

14. What challenges are associated with the effective use of AI in business analytics?

- a) No need for highly qualified specialists
- b) Ensuring a high level of cybersecurity and confidentiality
- c) No need for business process automation

- d) Using AI poses no challenges in the field of business analytics

APPENDIX 2

The literacy test

1. I understand the basic concepts of artificial Intelligence (AI).
2. I can explain the use of AI in various fields.
3. I have good skills in using intelligent systems.
4. I have a good understanding of the principles of artificial neural networks.
5. I have experience using genetic algorithms in AI.
6. I can identify problems that can be solved using AI.
7. I understand the ethical and confidentiality issues in AI.
8. I believe that AI can help solve complex tasks in various industries.
9. I believe that AI can improve the quality of life in society.
10. I have an idea of the role of AI in the development of business and the economy.
11. I can independently use AI to solve practical problems.
12. I believe that AI literacy is an important competence for modern education.
13. I understand how AI can impact the field of business analytics.
14. I have an understanding of the challenges and issues related to the use of AI in business analytics.
15. I believe that AI can help address global challenges facing humanity.

APPENDIX 3

The empowerment test

1. I consider myself capable of handling the AI literacy course effectively.
2. I believe that AI knowledge is significant for my education and professional development.
3. I trust that my AI skills can significantly impact my career.
4. My AI knowledge allows me to creatively express myself in professional activities.
5. I think working with AI enhances my overall effectiveness.
6. My AI skills influence my ability for self-improvement.
7. Integrating AI into my activities has improved my creative self-efficacy.
8. I believe that AI knowledge contributes to my personal and professional self-realization.

9. My AI knowledge is meaningful as it expands my possibilities.
10. I am confident that AI positively influences my ability for innovative thinking.
11. I consider the development of AI literacy crucial for the advancement of modern society.
12. My creative self-efficacy has improved due to the use of AI.
13. My AI skills positively impact my ability for self-organization.
14. AI knowledge has increased my self-productivity in learning and work.
15. My AI skills have boosted my self-esteem and confidence.
16. I believe that AI literacy development contributes to solving complex human problems.
17. Implementing AI allows me to feel my contribution to the development of the modern technological era.

APPENDIX 4

Semi-structured interview questionnaire: Students' perspectives on the artificial intelligence (AI) course

General questions:

How do you evaluate the overall organization and structure of the AI course?

What aspects of the course did you like the most?

In your opinion, which aspects of the course could be improved?

Teaching methods:

How do you assess the effectiveness of lectures in the course?

Were the practical sessions beneficial for you?

Did you find other teaching methods useful, and if so, which ones?

Assessment of understanding:

How would you rate your level of understanding of AI concepts after completing the course?

What difficulties did you encounter in studying the material?

Impact of the course:

How has studying AI influenced your perspective on this field?

Do you believe that the knowledge gained will be useful in the future?

Additional thoughts and observations:

Do you have any additional comments you would like to express regarding the course?

In your opinion, what would you like to learn more about or add to the course?

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